University of Botswana

Department of Mathematics

Tutorial 1: Limits

Part I: Evaluating Basic Limits

In Question 1–4, find the limit of the following functions by direct substitution where applicable.

1. $\lim\limits_{x \to 2} (\sqrt{9} - 5)$

2. $\lim\limits_{x \to 7} \dfrac{3x}{\sqrt{x+2}}$

3. $\lim\limits_{x \to 0} \sin(\dfrac{\pi + x}{2})$

4. $\lim\limits_{x \to 0} \sec 2x$

Part II: Evaluating Limits by Algebraic Simplification

In Question 5–9, use the appropriate factoring or dividing out technique to find the following limits.

5. $\lim\limits_{x \to 5} \dfrac{3x^3 - 15x^2}{x - 5}$

6. $\lim\limits_{x \to -4} \dfrac{x^3 + 64}{x + 4}$

7. $\lim\limits_{x \to 5} \dfrac{5 - x}{25 - x^2}$

8. $\lim\limits_{x \to 2} \dfrac{x^2 + 2x - 8}{x^2 - x - 2}$

9. $\lim\limits_{x \to 3} \dfrac{x^2 + x - 6}{x^2 - 9}$

Part III: Evaluating Limits by Rationalizing

In Question 10–14, use the rationalizing technique where applicable to find the following limits.

10. $\lim_{x \to 9} \dfrac{x - 9}{\sqrt{x} - 3}$

11. $\lim_{x \to 4} \dfrac{x - 2}{\sqrt{2x + 5} - 3}$

12. $\lim_{x \to 0} \dfrac{\sqrt{2 + x} - \sqrt{2}}{x}$

13. $\lim_{x \to 1} \dfrac{\sqrt{5x} - \sqrt{5}}{x - 1}$

14. $\lim_{x \to -1} \dfrac{2 - \sqrt{x + 5}}{x + 1}$

Part IV: Special Trigonometric Limits

In Question 15–23, use the appropriate trigonometric limit theorem to find the following limits.

15. $\lim_{x \to 0} \dfrac{2 \sin 9x}{6x}$

16. $\lim_{x \to 0} \dfrac{5 - 5 \cos x}{3x}$

17. $\lim_{x \to 0} \dfrac{\sin x(1 - \cos x)}{x^2}$

18. $\lim_{\theta \to 0} \dfrac{\cos \theta \tan \theta}{\theta}$

19. $\lim_{x \to 0} \dfrac{\sin^2 x}{x}$

20. $\lim_{x \to 0} \dfrac{\tan 7x}{3x}$

21. $\lim_{h \to 0} \dfrac{(1 - \cos h)^2}{h}$

22. $\lim_{t \to 0} \dfrac{\sin 3t}{2t}$

23. $\lim_{x \to 0} \dfrac{\sin 2x}{\sin 3x}$

Part V: Squeeze Theorem

In Question 24, use the Squeeze or Sandwich Theorem to find $\lim_{x\to 0} f(x)$.

24. State the sandwich theorem and hence use it to find

$$\lim_{x\to 0} \frac{1-\cos x}{x^2}$$

given that

$$\frac{1}{2} - \frac{x^2}{24} < \frac{1-\cos x}{x^2} < \frac{1}{2}.$$

Part VI: Piecewise Limits

In Question 25–26, find the following limits if they exist (if not explain why?).

25. Given

$$f(x) = \begin{cases} x^2, & x \leq 2 \\ 8-2x, & 2 < x < 4 \\ 4, & x \geq 4 \end{cases}$$

find $\lim_{x\to 2} f(x)$ and $\lim_{x\to 4} f(x)$

26. Determine $\lim_{x\to \pi} f(x)$, given

$$f(x) = \begin{cases} 1-\cos x, & x \leq \frac{\pi}{2} \\ 2\sin x, & x > \frac{\pi}{2} \end{cases}$$